

Cosmology & Astronomical Instrumentation

Problem Set — Solutions

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Section A — Objective

O1. Answer: (c). $\text{LGP} \propto D^2$, so the ratio is $(10/2)^2 = 25$.

O2. Answer: (b). A point source's image is always squeezed into essentially the same diffraction spot regardless of $f/\#$; only the raw collecting area D^2 matters. Focal ratio governs the brightness of *extended* images instead.

O3. Answer: (a). $\theta = 1.22\lambda/D = 1.22(550 \times 10^{-9})/1.0 \approx 6.71 \times 10^{-7} \text{ rad} \approx 0.14''$.

O4. Answer: (c) is false. Type Ia supernovae are explicitly calibrated *using* Cepheids observed in the same host galaxies, not independently. (a), (b), (d) are all correct.

O5. Answer: (a). $1 + z = 1/a_{\text{then}} = 1/0.25 = 4 \implies z = 3$.

Section B — Subjective

S1.

(i) $N_\star t = 500$; $n_{\text{pix}}(N_{\text{sky}}t + N_{\text{dark}}t + R^2) = 25(2000 + 200 + 64) = 56600$.

$$\text{SNR} = \frac{500}{\sqrt{500 + 56600}} \approx \frac{500}{239.0} \approx 2.09.$$

(ii) $N_\star t = 5000$; $n_{\text{pix}}(\dots) = 25(20000 + 2000 + 64) = 551600$.

$$\text{SNR} = \frac{5000}{\sqrt{5000 + 551600}} \approx \frac{5000}{746.1} \approx 6.70.$$

Going from $t = 100 \text{ s}$ to $t = 1000 \text{ s}$ ($10\times$ longer), SNR grew by a factor of $6.70/2.09 \approx 3.20 \approx \sqrt{10}$ — background-dominated growth, not linear. Indeed $N_\star t \ll n_{\text{pix}}(N_{\text{sky}} + N_{\text{dark}})t$ at both exposure times ($500 \ll 56600$, $5000 \ll 551600$), confirming the observation is deep in the **background-dominated** regime throughout.

(iii) Solving $\text{SNR}^2[(N_\star + n_{\text{pix}}(N_{\text{sky}} + N_{\text{dark}}))t + n_{\text{pix}}R^2] = N_\star^2 t^2$ with $\text{SNR} = 10$ gives the quadratic $25t^2 - 55500t - 160000 = 0$, whose positive root is

$$t \approx 2.2 \times 10^3 \text{ s} \approx 37 \text{ minutes}.$$

S2.

- (i) $\theta_{\min} = \lambda/B = (1.3 \times 10^{-3})/(8 \times 10^6) \approx 1.625 \times 10^{-10} \text{ rad} \approx 33.5 \mu''$.
- (ii) Single dish: $\theta = 1.22\lambda/D = 1.22(1.3 \times 10^{-3})/12 \approx 1.32 \times 10^{-4} \text{ rad} \approx 27.3''$. Improvement factor: $27.3''/33.5 \mu'' \approx 8.1 \times 10^5$ — roughly 800,000× finer resolution from interferometry than any single dish in the array could achieve alone.
- (iii) Since $\theta_{\min} \approx 33.5 \mu''$ is *smaller* than the shadow's $42 \mu''$ diameter, the baseline is just capable of resolving it — only about $42/33.5 \approx 1.25$ resolution elements fit across the shadow, meaning the real EHT was operating right at the edge of what its longest baselines could resolve, consistent with how marginal (though ultimately successful) the actual 2019 imaging was.

S3.

- (i) $v = cz = (3 \times 10^5 \text{ km/s})(0.05) = 15000 \text{ km/s} \implies d = v/H_0 = 15000/70 \approx 214 \text{ Mpc}$.
- (ii) $a_{\text{then}} = 1/(1+z) = 1/1.05 \approx 0.952$.
- (iii) $H_0 = 70 \text{ km/s/Mpc} = (7 \times 10^4)/(3.086 \times 10^{22}) \approx 2.27 \times 10^{-18} \text{ s}^{-1}$. Then

$$\rho_c = \frac{3H_0^2}{8\pi G} \approx \frac{3(2.27 \times 10^{-18})^2}{8\pi(6.674 \times 10^{-11})} \approx 9.2 \times 10^{-27} \text{ kg/m}^3,$$

which corresponds to $\rho_c/m_H \approx 5.5$ hydrogen atoms per cubic metre. Despite being almost inconceivably dilute — far emptier than any laboratory vacuum — this density is cosmologically decisive because it's the threshold multiplied across the *entire* volume of the universe: whether the true average density sits above, at, or below this tiny value determines whether the universe is closed, flat, or open.